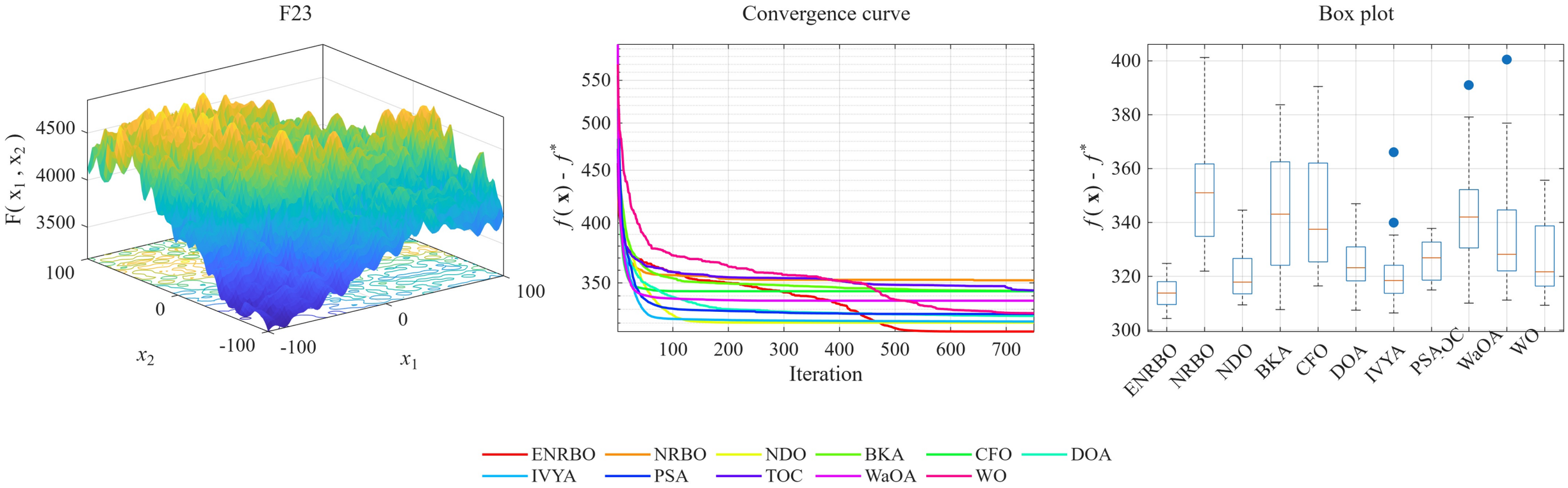

F23
F(x1 , x2)
4500
4000
3500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
550
500
450
400
350
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
400
380
360
340
320
300
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO